

9(a)	The volt is the potential difference between two points in a circuit in which one joule of electrical energy is converted to other forms of energy when one coulomb passes from one point to the other. Comment: - many could not explain the meaning of volt, i.e. one volt.	[1] for correct definition

9(b)	e.m.f. of a cell is the electrical potential energy converted from other forms of energy when unit charge flows through the cell. p.d. is the work done in converting electrical potential energy to other forms of energy when unit charge flows across two points in the circuit. <i>Comments:</i> - many showed inadequate understanding of energy transfer in distinguishing between e.m.f. and p.d.. - no appreciation that energy transfer is for unit amount of charge. - a common misconception was associating "a charge" to "a unit amount of charge".	[1] for correct explanation for e.m.f. [1] for correct explanation for p.d.
9(c)(i)	$I = \frac{V}{R} = \frac{2.0}{8.0} = 0.25 \text{ A}$ <i>Comments:</i> - many incorrectly the expression $P = IV$ to calculate the current, the concept of this approach is to calculate the current through the light bulb. - many did not realise the resistance of the potentiometer wire being half its full value.	[1] for correct equation [1] for correct answer
9(c)(ii)	$P = IV$ $I = \frac{0.50}{2.0} = 0.25 \text{ A}$ Total current through AX = $0.25 + 0.25 = 0.50 \text{ A}$ $V_{AX} = IR = (0.50)(8.0) = 4.0 \text{ V}$ <i>Comment:</i> - many did not recognise the current passing through the required section of the potentiometer wire was the sum of the currents from the light bulb and the first half of the potentiometer wire section.	[1] for correct current through bulb [1] for correct total current [1] for correct final answer
9(c)(iii)	$E = Ir + V$ $E = (0.50)(6.0) + 6.0 = 9.0 \text{ V}$ <i>Comment:</i> - misconceptions were evident as many candidates omitted the parallel resistance combination between the light bulb and the potentiometer wire when calculating the total resistance of the circuit.	[1] for correct equation [1] for correct answer
9(d)	The advantage of using the variable resistor is that a larger current can pass through the bulb to achieve greater brightness. The disadvantage is that the range of voltage that can be applied across the bulb is smaller. <i>Comment:</i> - many lack understanding of the application of series and parallel circuits in controlling the current through the light bulb.	[1] for valid advantage [1] for valid disadvantage

9(e)(i)	As the sliding contact X is adjusted in the direction of B, the potential difference (p.d.) across AX increases. There will be a point along AB where the p.d. across AX is equal to the e.m.f. of cell E_2. When this happens, no current will flow through E_2, hence giving zero galvanometer reading. <i>Comments:</i> - there is misconception of net flow of current being zero. - poor use of scientific language, such as potential difference. - many did not recognise E_2 being a cell, not the e.m.f. of a cell.	[1] for balanced potentials [1] for no current
9(e)(ii)	The balance length AX will be too small to be reliable since the percentage uncertainty will be large as the value of the balance length approaches the value of the uncertainty of the measurement of the balance length. <i>Comments:</i> - many mistook the balanced length would increase beyond the length of the potentiometer wire if the e.m.f. of E_1 is much larger than that of E_2. - many did not use precise scientific terms here, for example, mistakenly use accuracy instead of uncertainty or percentage error.	[1] for stating small balance length [1] for correct explanation
9(f)(i)	1.10 V	[1] for correct answer
9(f)(ii)	Resistor across balance length CY $R_{CY} = \frac{0.725}{1.000} \times 2.0$ $= 1.45 \, \Omega$ Total resistance across AD = R_{total} $V_{AB} = \frac{R_1}{R_{total}} \times V_{AD} \quad \text{--- (1)}$ $V_{CY} = \frac{R_{CY}}{R_{total}} \times V_{AD} \quad \text{--- (2)}$ $\frac{(2)}{(1)} \quad \frac{V_{CY}}{V_{AB}} = \frac{R_{CY}}{R_1}$ E.m.f. of thermocouple, $V_{CY} = \frac{R_{CY}}{R_1} \times V_{AB}$ $= \frac{1.45}{110} \times 1.10$ $= 0.0145 \, \text{V} \, (14.5 \, \text{mV})$ <i>Comment:</i>	[1] for correct R_{CY} [1] for correct expression of V_{AB} and V_{CY} [1] for correct answer

	<i>- many who thought that the resistance R_1 constituted part of the length of the potentiometer CD. They then used the potential divider formula to calculate the potential difference across CY.</i>	
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